Technical Report: Aerodynamic Limitations of the Diverging Ram Inlet

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The feasibility of the Diverging Ram Inlet concept was evaluated via analytical operator composition and 2D inviscid CFD simulations at Mach 5. Following an audit of initial modeling assumptions, this study examines the impact of previously overlooked Prandtl-Meyer expansion fans and geometric scaling. The findings demonstrate severe static pressure degradation, rendering it necessary to explore alternative inlet concepts for the design of unstart-mitigating hypersonic inlet systems.

I. Nomenclature

A	=	area
M	=	Mach number
P	=	static pressure
T	=	static temperature
$\mathbf{x}$	=	flow state vector
Y	=	Area-Mach relation
θ	=	flow turning angle
ρ	=	density
0	=	subscript - stagnation property
∞	=	subscript - freestream condition

II. Introduction

Achieving positive thrust from takeoff to hypersonic cruise without discrete mode transition remains a formidable challenge in aerospace engineering. Months prior to this writing, Wang Flight Systems Research Division LLC established an initial analytical framework investigating a mode-transition-free architecture combining a Diverging Ram Inlet (DRI), a rim-driven feathering compressor, and a Rotating Detonation Engine (RDE) (Figure 1). That initial study reported favorable theoretical metrics, including high stagnation pressure recovery and substantial net thrust. The organization then maintained an optimistic stance during a subsequent multi-month pause in technical operations.

Recently, shifting internal priorities to establish a more rigorously grounded research base prompted a formal technical audit of the architecture. This audit revealed complications in the fixed-geometry Diverging Ram Inlet concept, which harm its applicability in hypersonic propulsion. In alignment with the company's Open Science Commitment, which prioritizes research transparency and continuous self-correction, the company publicly reports these findings, establishing a corrected baseline for future development.

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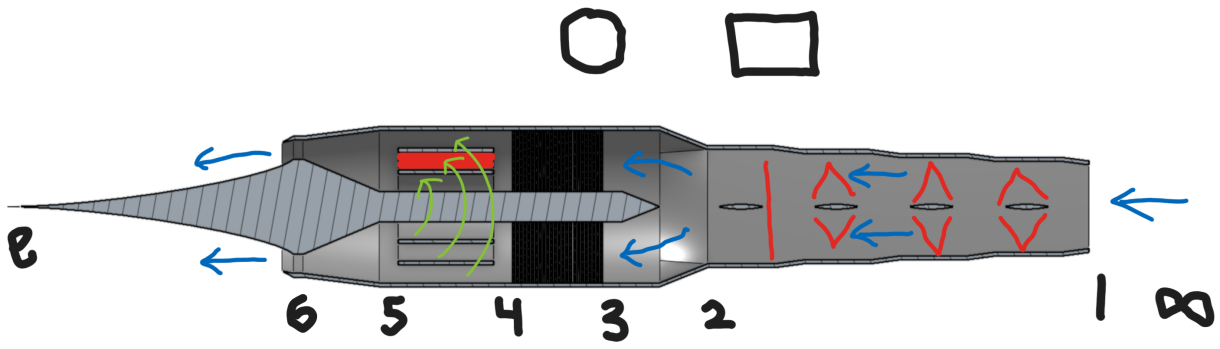


Fig. 1 Old engine flowpath schematic. Grey: walls. Black: turbomachinery. Blue: flowpath. red: shocks. green: detonation wave rotation direction. Not to scale.

III. Analytical Model Premises

The initial analytical framework modeled the Diverging Ram Inlet as a series of modules; each one consists of an isentropic diverging duct followed by incident and reflected oblique shocks at a diamond wedge on the centerline. The flow state vector $\mathbf{x} = [M, P, T, \rho]^T$ after a Diverging Ram Inlet was calculated via operator composition as

$$\mathbf{x_2} = \left(\mathbf{\Gamma}^{\theta_d}\,\mathbf{\Gamma}^{\theta_d}\,\mathbf{\Phi}_{\mp}^{H_N \leftarrow H_{N-1}}\,\mathbf{\Gamma}^{\theta_d}\,\mathbf{\Gamma}^{\theta_d}\,\mathbf{\Phi}_{\mp}^{H_{N-1} \leftarrow H_{N-2}} \cdots \mathbf{\Gamma}^{\theta_d}\,\mathbf{\Gamma}^{\theta_d}\,\mathbf{\Phi}_{\mp}^{H_2 \leftarrow H_1}\,\mathbf{\Gamma}^{\theta_d}\,\mathbf{\Gamma}^{\theta_d}\,\mathbf{\Phi}_{\mp}^{H_1 \leftarrow H_0}\right)\mathbf{x_1}. \tag{1}$$

The isentropic duct operator $\mathbf{\Phi}$ maps flow properties across a monotonic variable-area isentropic channel, with enforcement of a no-choke requirement:

$$\mathbf{x_2} = \mathbf{\Phi}_{\mp}^{A_2 \leftarrow A_1}\,\mathbf{x_1} = \begin{bmatrix} Y_{\mp}^{-1}\left(\dfrac{A_2}{A_1}\,Y(M_1)\right) \\ P_1 \cdot \dfrac{S_P(M_1)}{S_P(M_2)} \\ T_1 \cdot \dfrac{S_T(M_1)}{S_T(M_2)} \\ \rho_1 \cdot \dfrac{S_\rho(M_1)}{S_\rho(M_2)} \end{bmatrix}, \qquad A^* \notin (A_1, A_2) \tag{2}$$

The Area-Mach relation $Y(M)$ and it's subsonic/supersonic inverses Y_-^{-1} and Y_+^{-1} (see Figure 2) are defined as

$$\frac{A}{A^*} = Y(M) = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}}, \tag{3}$$

and the stagnation properties are defined as

$$S_T(M) = \frac{T_0}{T} = 1 + \frac{\gamma-1}{2}M^2 \qquad S_P(M) = \frac{P_0}{P} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{\gamma}{\gamma-1}} \qquad S_\rho(M) = \frac{\rho_0}{\rho} = \left(1 + \frac{\gamma-1}{2}M^2\right)^{\frac{1}{\gamma-1}} \tag{4}$$

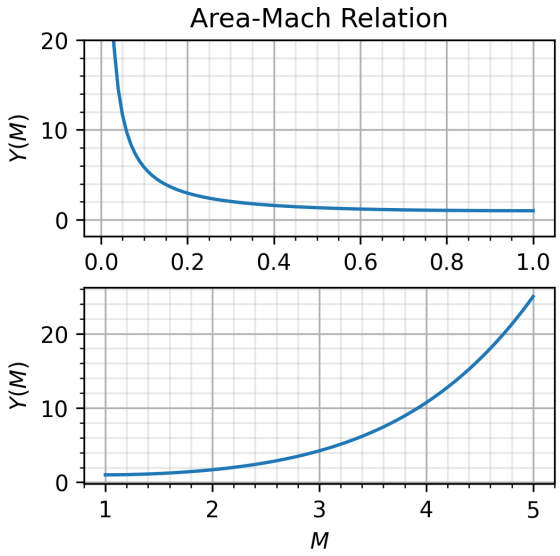


Fig. 2 Area-Mach relation $Y(M)$.

The normal shock operator $\mathbf{\Pi}$ is

$$\mathbf{x_2} = \mathbf{\Pi x_1} = \begin{bmatrix} M_2 = \sqrt{\frac{(\gamma-1)M_1^2+2}{2\gamma M_1^2-(\gamma-1)}} \\ P_1 \dfrac{2\gamma M_1^2 - (\gamma-1)}{\gamma+1} \\ T_1 \dfrac{[1+\frac{2\gamma}{\gamma+1}(M_1^2-1)](2+(\gamma-1)M_1^2)}{(\gamma+1)M_1^2} \\ \rho_1 \dfrac{(\gamma+1)M_1^2}{(\gamma-1)M_1^2+2} \end{bmatrix}, \qquad M_1 > 1. \tag{5}$$

For oblique shocks, the shock angle β_s can be calculated via

$$\tan\theta = \frac{2\cot\beta_s(M_1^2\sin^2\beta_s - 1)}{M_1^2(\gamma + \cos 2\beta_s) + 2}, \tag{6}$$

and the normal shock operator is applied to the normal Mach component, yielding

$$\begin{bmatrix} M_{2n} \\ P_2 \\ T_2 \\ \rho_2 \end{bmatrix} = \mathbf{\Pi} \begin{bmatrix} M_{1n} \\ P_1 \\ T_1 \\ \rho_1 \end{bmatrix}. \tag{7}$$

The total post-shock Mach number M_2 is recovered via

$$M_{2t} = M_{1t}\sqrt{\frac{T_1}{T_2}} \qquad M_2 = \sqrt{M_{2t}^2 + M_{2n}^2}. \tag{8}$$

The following steps define the Oblique Shock Operator, with respect to the turning angle limit:

$$\mathbf{x_2} = \mathbf{\Gamma}^{\theta}\,\mathbf{x_1} \tag{9}$$

$$\mathbf{\Gamma}^{\theta} = \begin{cases} \mathbf{I} & M_1 < 1 \\ \mathbf{\Pi} & \theta > \theta_{\max} \\ \text{[calculate above steps]} & \text{otherwise} \end{cases} \tag{10}$$

It is important to note that M_2 of the Oblique Shock Operator points in the direction of turning, not the horizontal direction. The previous hypothesis was that the diverging geometry avoided triggering the Kantrowitz limit, while the oblique shocks performed compression in hypersonic flow.

IV. Aerodynamic Limitations

The quasi-1D model did not model Prandtl-Meyer (PM) expansion fans, which occur when supersonic flow encounters a convex corner. The governing equation is the Prandtl-Meyer function,

$$\nu(M) = \int \frac{\sqrt{M^2-1}}{1+\frac{\gamma-1}{2}M^2}\frac{dM}{M} \tag{11}$$

$$= \sqrt{\frac{\gamma+1}{\gamma-1}}\arctan\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} - \arctan\sqrt{M^2-1}. \tag{12}$$

The post-expansion Mach number in the rotated direction is calculated by

$$\nu(M_2) = \nu(M_1) + \theta, \tag{13}$$

and stagnation properties are conserved, yielding the expansion fan operator,

$$\mathbf{x_2} = \mathbf{\Xi}^{\theta}\mathbf{x_1} = \begin{bmatrix} \nu^{-1}(\nu(M_1)+\theta \\ P_1\frac{S_P(M_1)}{S_P(M_2)} \\ T_1\frac{S_T(M_1)}{S_T(M_2)} \\ \rho_1\frac{S_\rho(M_1)}{S_\rho(M_2)} \end{bmatrix}, \qquad \theta < \theta_{\max} \tag{14}$$

These analytical operators demonstrate how PM expansion fans destroy static pressure in Diverging Ram Inlet module variational architectures (1). All runs were done with $M_\infty = 5$ at 30000 m altitude and exclude shock reflections.

Table 1 Analytical operator chains and static pressure variations for Diverging Ram Inlet configurations.

Variation	Description	Operator Chain	Static Pressure Ratio
1	12° convex corner and matching compression corner, same wall	$\Gamma^{12^\circ}\Xi^{12^\circ}\mathbf{x}_\infty$	0.9289
2	6° convex corner, 18° compression corner, 12° convex corner, same wall	$\Xi^{12^\circ}\Gamma^{18^\circ}\Xi^{6^\circ}\mathbf{x}_\infty$	1.0034
3	Triangular wedge, 15° incident angle	$\Gamma^{15^\circ}\Xi^{30^\circ}\Gamma^{15^\circ}\mathbf{x}_\infty$	0.9116

The previous paper also argued that the Diverging Ram Inlet is immune to subsonic choking, because the diverging duct acts as a diffuser driving flow away from Mach 1. That argument is inaccurate given the mathematical realities of elliptic subsonic flow. In subsonic flow, pressure characteristics may travel upstream. If the end of the diverging section exhibits a critically low pressure, the diffuser demands an adverse pressure gradient, forcing even lower pressure at the inlet face. The pressure difference between the freestream and what is demanded of the inlet face drives the flow to choke as streamlines bend to enter the narrow inlet, as if it were an invisible giant converging duct.

V. CFD Findings and Anomalies

Inviscid CFD using 2D `rhoCentralFoam` was performed on the three variational architectures at $M_\infty = 5$ and an altitude of 30,000 m. Flow moves from right to left. Grid refinement utilized a cell-multiplying `scale` parameter across levels 1, 2, and 3. Averaged static pressure ratios (not including reflections) for the three variational architectures are presented in Table 2.

Table 2 Cell count (top) and static pressure ratio (bottom) of Diverging Ram Inlet Architectures.

Architecture	Scale		
	1	2	3
1	11088	44352	99792
	0.262	0.699	0.779
2	11088	44352	99792
	2.683	3.674	3.103
3	12256	49024	110304
	0.737	0.685	0.685

Architectures 1 and 3 confirm static pressure collapse ($P/P_\infty < 1$) as shown in figures 3 and 5. While the reflections may allow for a marginal static pressure rise, it requires extended axial length. Architecture 2 presents a non-obvious result, where the 18° corner's shock distorts the subsequent expansion fan in a "quenching" effect. This causes the static pressure to rise. However, capturing this pressure gain requires excessive axial length requirements.

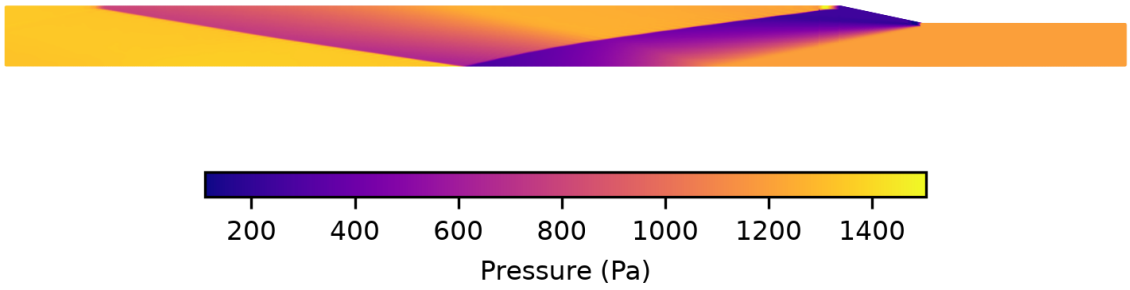


Fig. 3 Pressure Contour of Architecture 1.

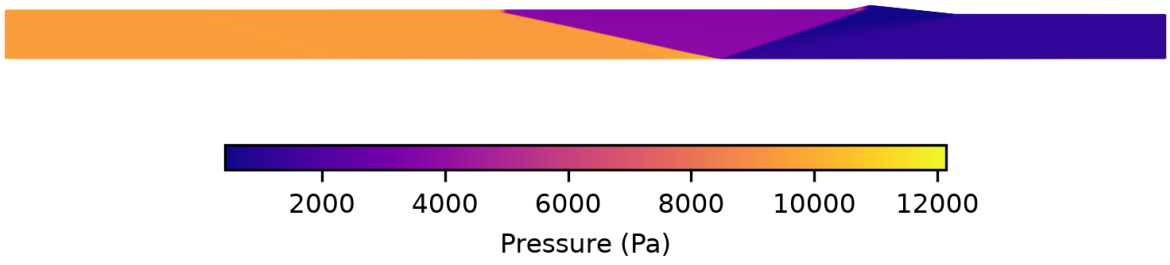


Fig. 4 Pressure Contour of Architecture 2.

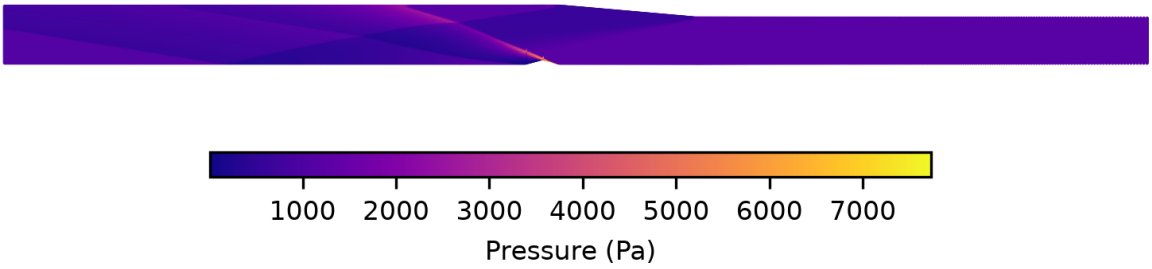


Fig. 5 Pressure Contour of Architecture 3.

Varying the scale parameter indicates an initial stabilizing trend within the CFD data. However, higher resolution scales are required to achieve full numerical convergence, which would necessitate runtime durations exceeding 24 hours. Pursuing this is deemed out of scope, as it would merely reaffirm the limitations of a concept the team is already moving away from. The quenching effect observed in Architecture 2 might warrant interest in future investigations to determine whether it represents genuine fluid dynamics or a numerical artifact that somehow survived mesh refinement.

VI. Conclusion

Analytical operator modeling and 2D CFD simulations demonstrate that the fixed-geometry Diverging Ram Inlet concept is infeasible for hypersonic propulsion due to aerodynamic pressure collapse and unacceptable length penalties. Wang Flight Systems Research Division LLC concludes its investigation of the DRI concept, establishing a corrected analytical baseline for future engine development.

Acknowledgments

None.

References

None.